

# Innovative Solutions

*In collaboration with other students, identify a problem, need or opportunity to analyse, design, develop and evaluate an innovative solution.*

## The big picture

This is a team, design-thinking outcome. You start from a driver of innovation — curiosity, ingenuity or a UN Sustainable Development Goal — examine what digital systems can do and how emerging technologies affect people, then run a full design process: collect user needs, ideate, design with mock-ups/storyboards/pseudocode, develop, document progress with Gantt charts and version control, and test and evaluate. Threaded through it is the law and ethics of emerging technology and AI. The emphasis is on collaboration, user-centred design and honest evaluation.

## How it's assessed

Expect scenario questions about what makes a solution innovative, choosing user-need collection methods and justifying them against constraints, naming the right legislation or AI Ethics Principle for a situation, and evaluating a solution's efficiency and effectiveness against criteria. Project-management items (Gantt charts, version control, journals) and ethics items are common. Generic "it helps people" answers score poorly — name the method, the law or the criterion.

## Key knowledge at a glance

| KK   | Topic                                                                   | In one line                                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KK01 | U2·O1 KK01 · Factors that inspire and drive innovation — Digital Vector | Every digital solution you'll ever study started as a spark in someone's head. This dot point is about that spark — the factors that inspire and drive innovation. Get this right and the rest of the Innovative Solutions outcome clicks into place.    |
| KK02 | Functions & capabilities of digital systems                             | Before you can design something innovative, you need a clear-eyed picture of what today's digital systems already do — and how much they are already able to do. This dot point is the map of the territory you're about to push past.                   |
| KK03 | The impact of emerging technologies                                     | Every emerging technology changes the world it lands in — and almost never in just one direction. The same system that lifts one person up can leave another behind. This module is about naming those impacts precisely, the way the exam wants you to. |

| KK          | Topic                                                              | In one line                                                                                                                                                                                                                                                                                                                      |
|-------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KK04</b> | What makes a solution stand out ?                                  | Anyone can build an app. Far fewer build one people actually adopt. This dot point is about the qualities that separate a forgettable project from a creative and innovative solution — and how to argue, in exam conditions, that a solution has them.                                                                          |
| <b>KK05</b> | Solution specifications what to build, how well, and where to stop | Before a single screen gets designed, an innovator writes the spec : the functional and non-functional requirements , the constraints they must work inside, and the scope that draws the line around the whole thing. Get this right and the build almost designs itself. Get it vague and the project drowns.                  |
| <b>KK06</b> | Before you build it, ask the people who'll use it.                 | An innovative solution that nobody needs is just an expensive hobby. This module is about the two techniques the study design names for finding out what users actually need — surveys and interviews — when to reach for each, and how to turn what people tell you into real requirements.                                     |
| <b>KK07</b> | Design tools: drawing the solution before you build it             | Four ways to put a design on paper — mock-ups, sitemaps, storyboards and pseudocode. Same solution, four different lenses. Knowing which lens shows what is the whole game in the exam.                                                                                                                                          |
| <b>KK08</b> | One great idea is the last idea you keep — not the first you have. | Meet Team Vector — Jade, Marcus, Priya and Tom, four Year 11 students at a Melbourne school. They've spotted an opportunity : new Year 7 students arrive lost, anxious, and afraid to ask for help in their first weeks. Team Vector decides to develop an innovative digital solution. The question is how — and "we'll just th |
| <b>KK09</b> | Documenting progress                                               | Straight from the VCAA Applied Computing Study Design (Unit 2, Area of Study 1), the examinable wording is:                                                                                                                                                                                                                      |
| <b>KK10</b> | Testing & evaluating an innovative solution                        | You built the thing. It runs. Now the hardest question in the whole project: does it actually work, and is it actually any good? Those are two different questions — and mixing them up is the single most common way to lose marks in this outcome.                                                                             |
| <b>KK11</b> | Legislation & frameworks                                           | It's tempting to think of legislation as a handbrake on creativity. The opposite is closer to the truth: rules are what let a clever prototype become something people will actually trust with their data, their health and their kids.                                                                                         |
| <b>KK12</b> | The ethics of emerging technology                                  | When new tech can do something, should it? This dot point is about the line between can and should — through cyber-security threats, biometric data, and the jobs machines replace.                                                                                                                                              |
| <b>KK13</b> | The ethics of artificial intelligence                              | Same tool. Three different ethical problems. Keep Priya's team in mind — every concept below maps onto something that happened to MuseBot.                                                                                                                                                                                       |

| KK          | Topic                                                       | In one line                                                                                                                                                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KK14</b> | Did it actually work?<br>Evaluating an innovative solution. | Building a solution proves you can. Evaluating it answers the harder question: was it worth it? This module is about judging a finished solution on two separate scales — effectiveness (how well it does the job) and efficiency (how little it wastes doing it) — using criteria you wrote before you built, and evidence you |

## Chapter glossary

| Term                                 | Meaning                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Innovation</b>                    | A solution that is original, effective and often disruptive — new value, not just newness. |
| <b>User-centred design</b>           | Designing around real user needs gathered and tested throughout the process.               |
| <b>Sustainable Development Goals</b> | UN goals that can inspire and drive innovative digital solutions.                          |
| <b>Gantt chart</b>                   | A timeline chart for planning tasks, durations and dependencies in a project.              |
| <b>Version control</b>               | Tracking and managing changes to a solution so work can be coordinated and recovered.      |
| <b>User testing</b>                  | Having intended users trial a solution to evaluate it.                                     |
| <b>AI Ethics Principles</b>          | Australia's voluntary principles guiding responsible AI design and use.                    |
| <b>Efficiency</b>                    | How economically a solution uses resources such as time, cost and effort.                  |
| <b>Effectiveness</b>                 | How well a solution meets its requirements and the user's need.                            |

## Answering across the chapter

These scenario questions pull together several key-knowledge points at once — the kind of integrated thinking the exam rewards. Compare the weak and strong responses.

## Q1. Evaluate · 4 marks

A student team builds an app that uses AI to recommend study resources. Evaluate the solution against efficiency and effectiveness, and identify one ethical issue they must manage.

### ✗ A weak answer

The app is good and helps students study, but AI can be biased. (→ *No criteria applied, ethics undeveloped — 1.*)

### ✓ A strong answer

**Effectiveness:** judge whether the recommendations actually match each student's needs and improve their study — tested through user testing. **Efficiency:** judge whether it delivers those recommendations quickly and without excessive data or cost. **Ethical issue:** the AI may produce **biased recommendations** if its training data is skewed, so under the AI Ethics Principles the team should test for fairness and be transparent about how recommendations are generated.

### How the marks are awarded

- 1 mark — applies effectiveness to the scenario.
- 1 mark — applies efficiency to the scenario.
- 1 mark — names a valid ethical issue (bias/transparency/privacy).
- 1 mark — links it to managing the issue (e.g. AI Ethics Principles).

## Q2. Justify · 3 marks

Justify using both a Gantt chart and version control when several students collaborate on one solution.

### ✗ A weak answer

They help the team stay organised and not lose work. (→ *Right idea, but each tool's distinct role isn't explained — 1-2.*)

### ✓ A strong answer

A **Gantt chart** plans who does what and by when, showing task durations and dependencies so the team can delegate and track progress against deadlines. **Version control** manages changes to the actual files, letting members work in parallel, merge contributions and roll back mistakes. One coordinates the schedule; the other protects the work — together they make collaboration reliable.

### How the marks are awarded

- 1 mark — Gantt chart = scheduling/delegation/dependencies.
- 1 mark — version control = managing changes/parallel work/recovery.
- 1 mark — distinguishes the two roles.

## Chapter revision checklist

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- Explain a driver of innovation (curiosity, ingenuity, an SDG).
  - Describe capabilities of digital systems and impacts of emerging technologies.
  - Identify characteristics of creative and innovative solutions.
  - Collect user needs with interviews/surveys and justify the choice.
  - Design with mock-ups, pseudocode, sitemaps and storyboards.
  - Document progress with Gantt charts, journals and version control.
  - Test and evaluate with user testing and expert review.
  - Apply legislation and the AI Ethics Principles to emerging-tech scenarios.